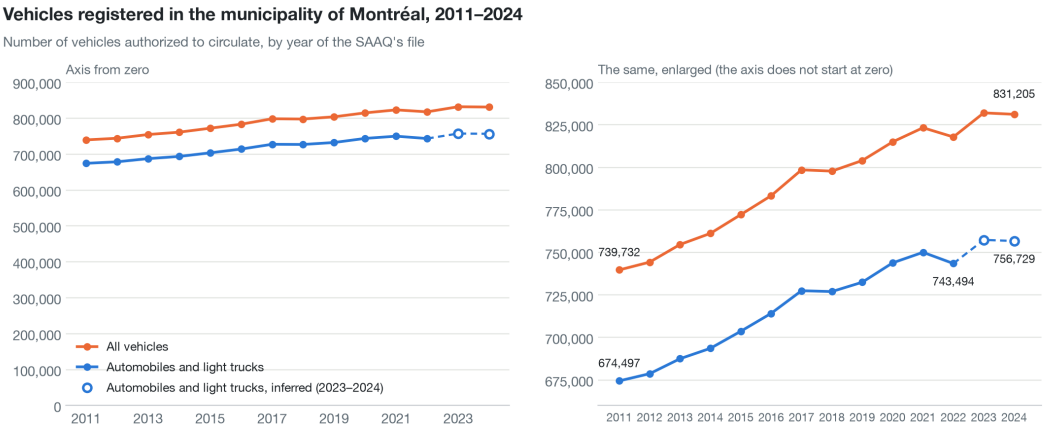
# Vehicle registrations in Québec, 2011–2024

## Joining the SAAQ's open data with the files released under the access Act

Rick Hoge, PhD

Draft, November 11, 2026 v2026-11-11-07-05-31

# Overview: the municipality of Montréal, 2011–2024




Type is not in the 2023–2024 files: it is inferred from make, model and model year. Applied to each published year 2012–2022 from the years before it, the method gives +0.05 % to +0.13 % against the published count (21 tests).
Vehicles left unassigned: 2,432 (2023), 2,917 (2024).
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# Two sources, one row per registered vehicle

| Years | Source | Columns |
|---|---|---|
| 2011–2022 | SAAQ open data, « Véhicules en circulation » (CC BY 4.0), files as revised by the SAAQ on December 18, 2023 | class, type, make, model, model year, net mass, cylinders, displacement, axles, colour, fuel (from 2017), region, MRC, municipality code |
| 2023–2024 | Files released by the SAAQ under the access Act, file 03.06.34729 (decision of September 17, 2025) | year, make, model, model year, municipality name, region, first three characters of the postal code (FSA) |

- Rows: 5,985,463 in 2011, 7,058,605 in 2022, 7,169,989 in 2023, 7,189,594 in 2024.
- The portal holds no year after 2022. Its note: « Les données ont été révisées pour respecter les nouvelles dispositions de la loi 25 ».
- The decision withheld the other columns under section 53 of the Act.

# What the two sources share

- **Shared:** make, model, model year, and the municipality.
- **Only in 2011–2022:** class, type, fuel, net mass. A type or fuel figure for 2023–2024 is therefore **inferred**, and is labelled so wherever it appears.
- **Only in 2023–2024:** the FSA. A figure by FSA exists for two years only.
- **Not inferred for 2023–2024:** the class (personal or commercial use). It describes the use made of a vehicle, not the vehicle, so make, model and model year do not settle it.

# Step 1: one vocabulary of makes and models

- The open data write makes and models as codes of at most five characters (TOYOT, COROL). The release mixes those with full names, other abbreviations and misspellings: more than 7,000 make strings against 418 codes.
- As released, 80.9 % of the 2023 vehicles and 75.6 % of the 2024 vehicles carry a make and model in the open data's own main spelling.
- Stated rules (first five characters, punctuation, two short alias tables, unambiguous prefixes, trim suffixes) bring this to 97.1 % and 96.8 %. A further 2.4 % and 2.7 % resolve to the make only; 0.4 % and 0.5 % stay unresolved. Nothing is matched by similarity.
- Some codes mean other vehicles in the release than in the open data (DODGE CARAV, merged into GRAND by the curators). They are detected from vehicles of model year 2021 or earlier, whose number per code cannot grow after 2022; 52 codes (148,434 vehicles in 2023) are reassigned by model-year profile.
- Municipality names are linked to the codes of 2011–2022 for 98.3 % of the 2024 vehicles (ISQ geographic codes, July 31, 2025).

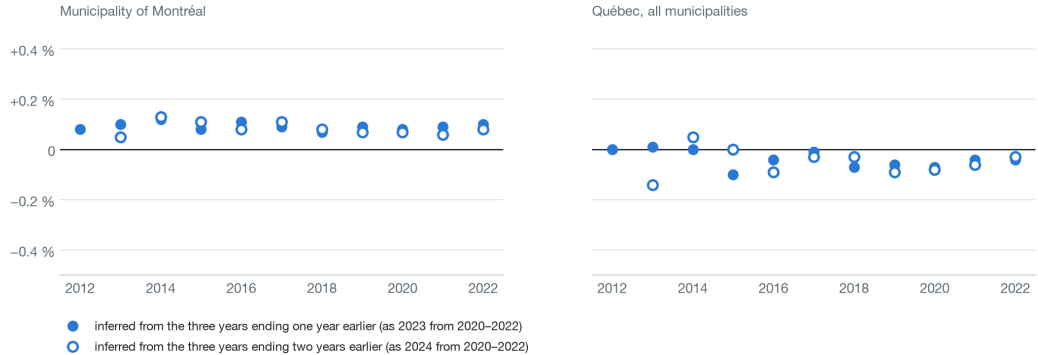
# Step 2: type and fuel as expected counts

- The open data of 2020–2022 give, for each make, model and model year, how many vehicles are of each type and each fuel.
- A vehicle of 2023–2024 is not given one value. It contributes to each value in the proportions observed for its key.
- The key is the most specific one found: make, model and model year (94.8 % of the 2023 vehicles, 88.3 % of 2024); then make and model on recent model years (2.0 %, 8.2 %); then make and model, make and model year, make.
- Vehicles matching no key stay unassigned: 0.5 % in 2023, 0.5 % in 2024.
- **The method is tested where the answer is known:** each published year is reduced to the release's seven columns, inferred from earlier years, and compared with what the SAAQ published.

# The test on published years: vehicle type

**Inferring vehicle type from make, model and model year: error on published years**

Automobiles and light trucks: inferred count against the count the SAAQ published, in per cent




Each published year is reduced to the seven columns of the 2023–2024 files, inferred, and compared with the published type. The test uses the open data's own make and model codes; it does not measure the effect of the unedited codes of 2023–2024.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# The test on published years: what it shows

| Quantity inferred | Difference from the published count |
|---|---|
| Automobiles and light trucks, Montréal (21 tests) | $+0.05$ % to $+0.13$ % |
| Automobiles and light trucks, Québec | $-0.14$ % to $+0.05$ % |
| Same, by municipality, 2022: under 1,000 vehicles | median 0.58 % |
| Same, by municipality, 2022: 10,000 to 99,999 | median 0.11 % |
| Gasoline, Québec, one year ahead | $-0.11$ % to 0.11 % |
| Electric, Québec, one year ahead | $-7.7$ % to $-3.8$ % |
| Electric, Québec, two years ahead | $-18.8$ % to $-12.4$ % |
| Plug-in hybrid, Québec, one year ahead | $-25.0$ % to $+11.1$ % |

- Type is recovered to a fraction of one per cent.
- Electric vehicles are under-counted, because the newest model years are missing from the reference: the inferred electric counts for 2023–2024 run low.
- The test uses the open data's own codes. It does not measure the effect of the release's unedited codes.

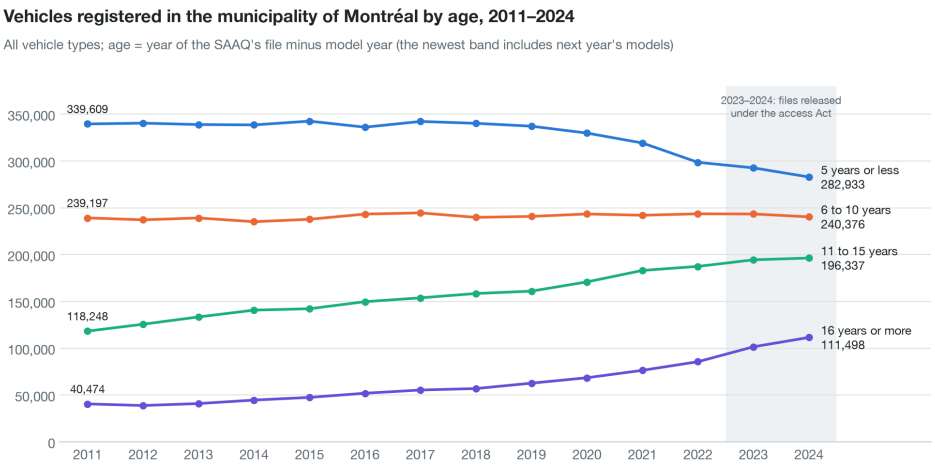
# Result: Montréal, automobiles and light trucks

| Municipality of Montréal | 2011 | 2021 | 2022 | 2023 | 2024 |
|---|---|---|---|---|---|
| All vehicles (released) | 739,732 | 823,314 | 817,880 | 832,054 | 831,205 |
| Automobiles and light trucks | 674,497 | 750,071 | 743,494 | *757,376* | *756,729* |

Italic: inferred. Released: SAAQ open data (files of December 18, 2023) to 2022; access release 03.06.34729 for 2023–2024.

- 2011 to 2021: 75,574 more automobiles and light trucks ($+11.2$ %).
- 2022 is lower than 2021 by 6,577 in the published files; all vehicles fall by 5,434 the same year.
- 2023 and 2024 (inferred) stand 7,305 and 6,658 above 2021; the two years differ by 647.

# Vehicles by age: nothing inferred

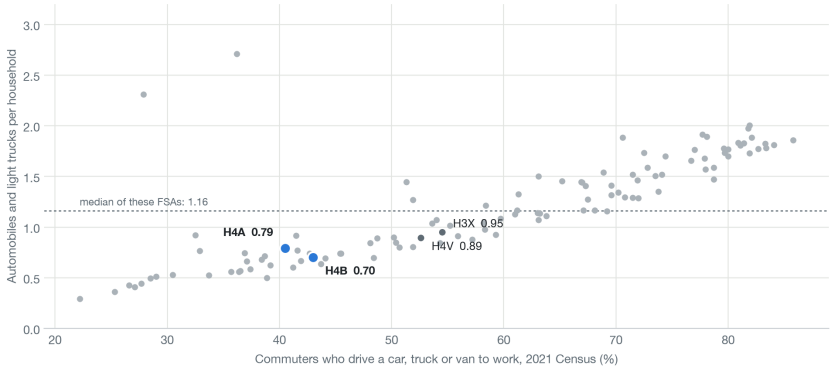



Nothing is inferred: both sources carry the model year. Vehicles with no usable model year: 58 to 2,764 a year, not shown.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# Automobiles per household by FSA, 2024

**Automobiles and light trucks per household, by FSA: island of Montréal and Laval, 2024**

Each dot is one FSA beginning with H (114 with 1,000 households or more); rue de Terrebonne runs through H4A and H4B




Vehicles: 2024 file, type inferred from make, model and model year; all owners, businesses included, so an FSA with fleets or dealers stands high. Households and commuting: 2021 Census profile by FSA. The two quantities are three years apart. Above the scale: H9P 3.49, H3B 6.32. Statistics Canada, 2021 Census, catalogue 98-401-X2021013.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# The FSAs around rue de Terrebonne

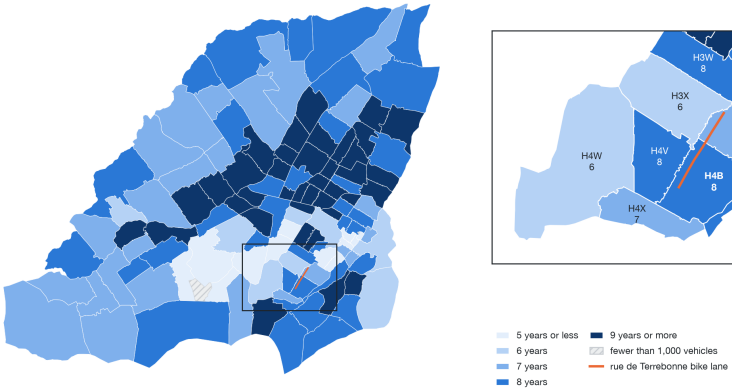
| FSA | Automobiles and light trucks, 2024 (inferred) | Households, 2021 Census | Per household | Commuters who drive (%) | Rank from the lowest, FSAs in H |
|---|---|---|---|---|---|
| H4A | 9,598 | 12,124 | 0.79 | 40.5 | 31 of 114 |
| H4B | 7,351 | 10,524 | 0.70 | 43.0 | 23 of 114 |
| H4V | 8,199 | 9,179 | 0.89 | 52.6 | 39 of 114 |
| H3X | 9,999 | 10,526 | 0.95 | 54.5 | 45 of 114 |

- Rue de Terrebonne runs through H4A and H4B. The median of the 114 FSAs is 1.16 per household.
- The vehicle count includes vehicles owned by businesses. The ratio is an average: it does not say how many households have no vehicle.
- Households are those of Statistics Canada's 2021 Census profile by FSA; vehicles are of 2024.

# Median vehicle age by FSA, 2024

**Median age of registered vehicles, by FSA, 2024**

Island of Montréal and Laval (postal codes beginning with H); all vehicle types; age = 2024 minus model year

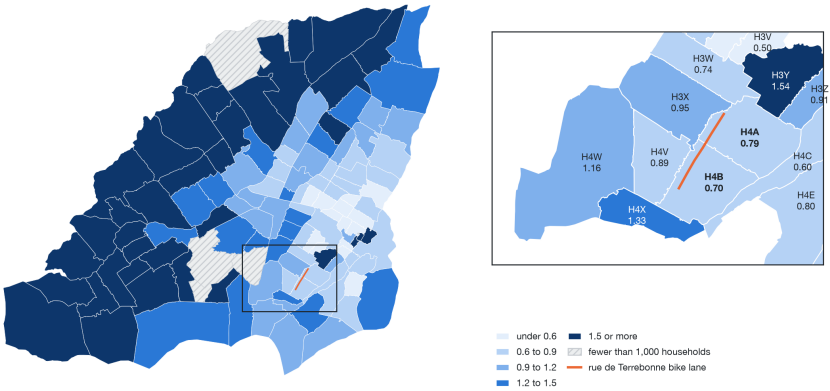



Nothing is inferred: the 2024 file carries the model year and the FSA. All owners, businesses included. FSA boundaries: DMTI Spatial, CanMap Postal Code Suite v2024.3.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# Automobiles per household by FSA, 2024: the map

**Automobiles and light trucks per household, by FSA, 2024**

Island of Montréal and Laval (postal codes beginning with H)

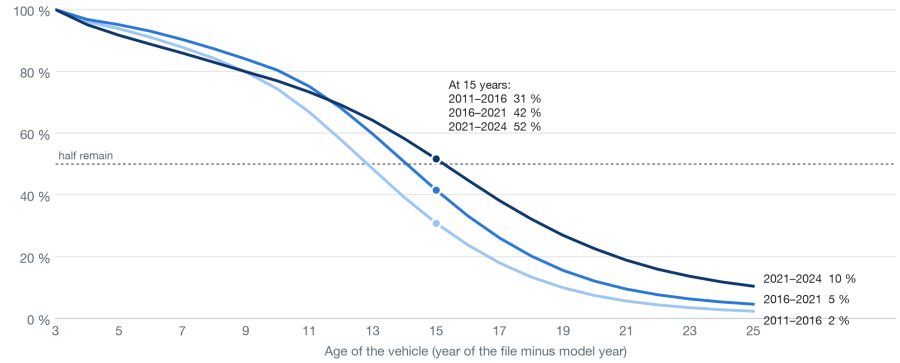



Vehicles: 2024 file, type inferred from make, model and model year; all owners, businesses included. Households: 2021 Census profile by FSA. FSA boundaries: DMTI Spatial, CanMap Postal Code Suite v2024.3. Statistics Canada, catalogue 98-401-X2021013.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# How long vehicles stay registered

**How long automobiles and light trucks stay registered in Québec**

Of the vehicles of a model year registered at age 3, the share still registered at each later age, by period




Each step compares one model year's count in two consecutive yearly files. Counts for 2023 and 2024 are inferred (type from make, model and model year), so the last period carries that method's error, under 0.2 % on published years. A vehicle that leaves Québec counts as a departure; one that arrives used counts as an arrival.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

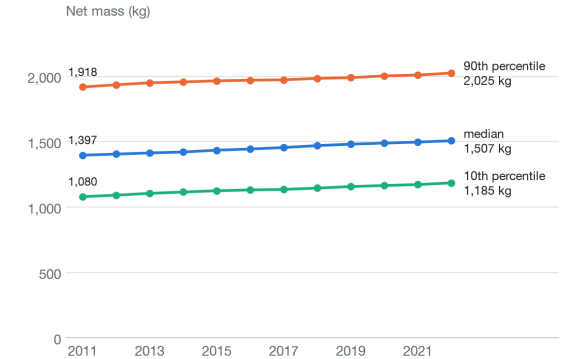
# An older fleet, kept longer

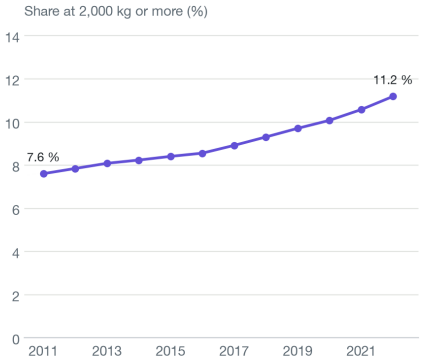
- Of the automobiles and light trucks registered in Québec at age 3, half are still registered at 13 years in 2011–2016, at 15 years in 2016–2021 and at 16 years in 2021–2024.
- At 15 years: 31 %, 42 % and 52 % remain in the three periods.
- In Montréal the mean age of automobiles and light trucks goes from 6.6 years in 2011 to 8.2 in 2022 (published) and 8.7 in 2024 (inferred); the share aged 10 years or more from 26 % to 42 %.
- The last period includes 2023 and 2024, whose counts by type are inferred. A vehicle that leaves Québec counts as a departure.

# Net mass, 2011–2022

**Net mass of automobiles and light trucks registered in Montréal, 2011–2022**

Municipality of Montréal; net mass is the vehicle empty, as recorded by the SAAQ



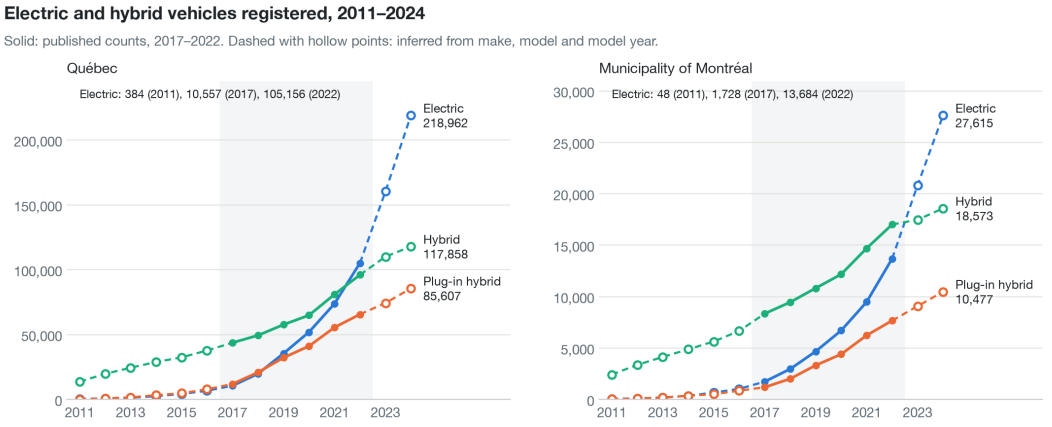




Mean 1,458 kg in 2011, 1,559 kg in 2022. The 2023–2024 files carry no mass. Mass is recorded for more than 99.9 % of these vehicles each year.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# Mass to the fourth power: what the index measures

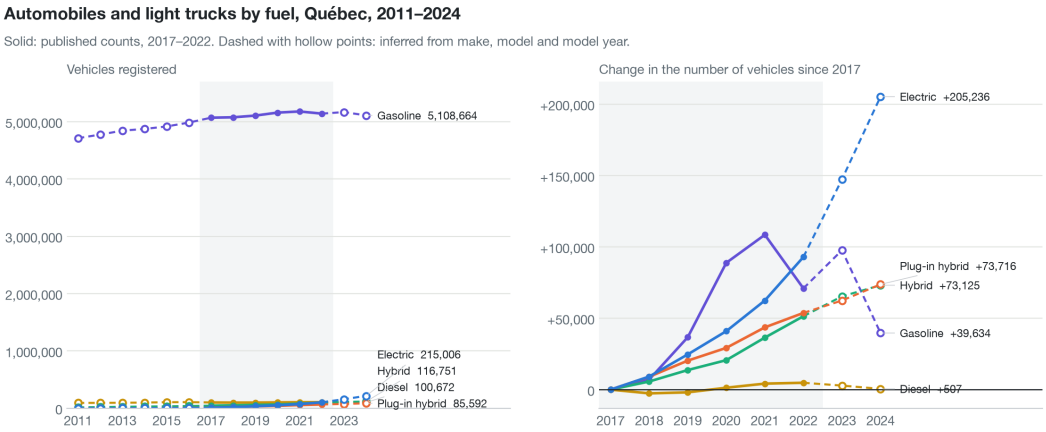
- For automobiles and light trucks in Montréal, the sum of net mass to the fourth power is 1.35 times its 2011 value in 2022, and the mean per vehicle 1.22 times. These reproduce the 2025 application's chart (1.35 and 1.22).
- The mean net mass goes from 1,458 kg to 1,559 kg; the share at 2,000 kg or more from 7.6 % to 11.2 %.
- Across automobiles, trucks, buses and tool vehicles together, automobiles are 95 % of the vehicles and 1.0 % of the summed fourth power in 2022. An all-type index follows a few tens of thousands of heavy vehicles.
- Its one-year drop in 2012 is in the files: 3,465 buses registered in Montréal in the 2011 file, 3,037 in 2012, 3,290 in 2013.
- The quantity uses the net mass (the vehicle empty) and no distance driven. It describes the registered fleet, not the wear of a street.

# Electric and hybrid vehicles, 2011–2024 (published 2017–2022, inferred before and after)




The fuel column begins in 2017, and the files of 2023–2024 carry none. Tested on published years for Québec, the inference forward (2023–2024) differs from the published count by −19 % to −4 % for electric, −44 % to +11 % for plug-in hybrid and −18 % to +5 % for hybrid, mostly low; backward (before 2017), tested up to three years back, by −2 % to +4 %, +1 % to +6 % and −1 % to +0 %. 2011–2016 lie up to six years before the files used, beyond what the test covers. Each panel has its own scale, from zero.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# Automobiles and light trucks by fuel: Québec

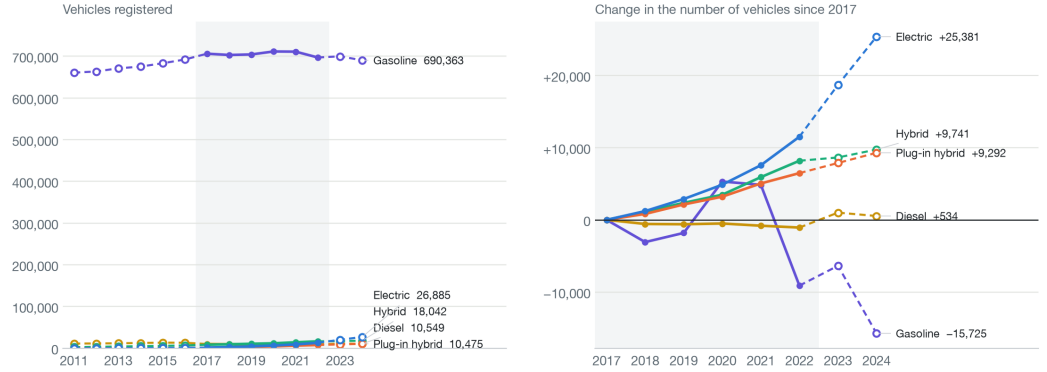



Type AU only (automobiles and light trucks), businesses included. Before 2017 the type is published and the fuel inferred; for 2023–2024 both are inferred. On published years, all vehicle types together, the forward inference is within 0.5 % for gasoline and runs 4 to 19 % low for electric; the backward one is within 6 %. Other fuels (propane, natural gas, not stated) are left out.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# Automobiles and light trucks by fuel: Montréal

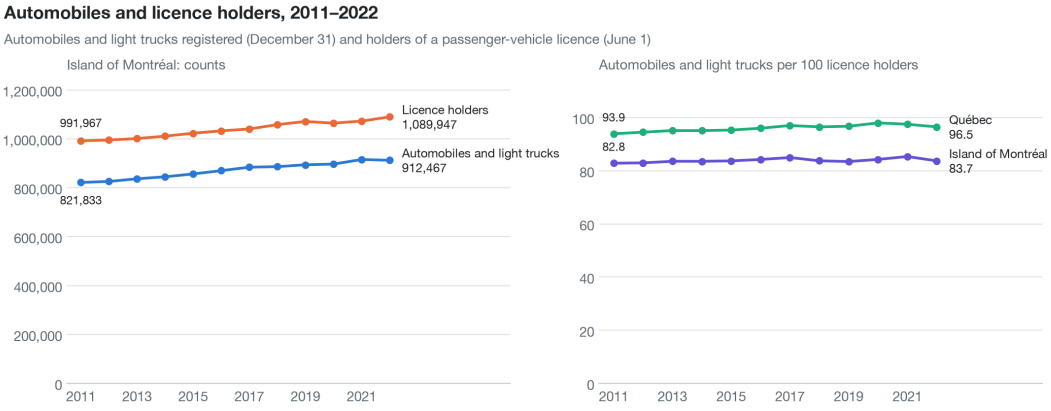
**Automobiles and light trucks by fuel, municipality of Montréal, 2011–2024**

Solid: published counts, 2017–2022. Dashed with hollow points: inferred from make, model and model year.




Type AU only (automobiles and light trucks), businesses included. Before 2017 the type is published and the fuel inferred; for 2023–2024 both are inferred. On published years, all vehicle types together, the forward inference is within 0.5 % for gasoline and runs 4 to 19 % low for electric; the backward one is within 6 %. Other fuels (propane, natural gas, not stated) are left out.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# Automobiles and licence holders, 2011–2022

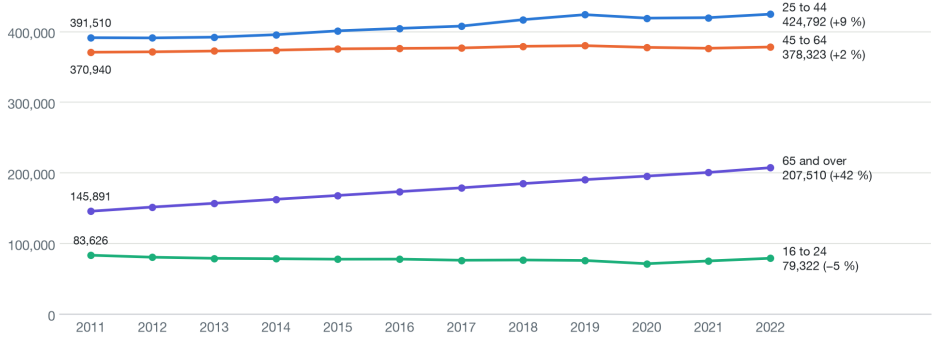



Licence holders: regular class 5 or probationary licence (the files give no class for probationary licences). The licence files give the MRC, not the municipality: the island is the City of Montréal and the fifteen other municipalities. Vehicles include those owned by businesses. Both series end in 2022. SAAQ, open data « Permis de conduire », 2011–2022.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# Licence holders by age, island of Montréal

**Licence holders on the island of Montréal by age, 2011–2022**

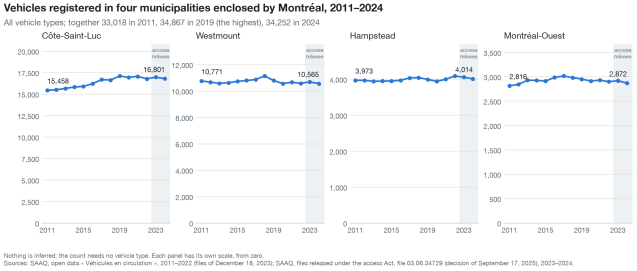
Holders of a passenger-vehicle licence (regular class 5 or probationary) on June 1, by age group




Counts of holders, not rates: no population is used. The licence files give the MRC, not the municipality: the island is the City of Montréal and the fifteen other municipalities. SAAQ, open data « Permis de conduire », 2011–2022.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# Other series checked against the 2025 application

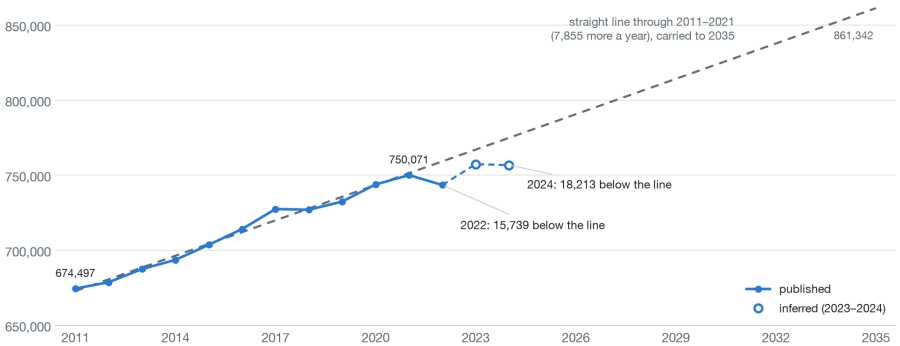
- 12 legend values of the 2025 charts were recomputed: 11 are identical. The other differs by 58 vehicles whose model year is invalid, which the application counted as 16 years or more.
- Hampstead, Montréal-Ouest, Côte-Saint-Luc and Westmount together: 33,018 vehicles in 2011, 34,867 in 2019, 34,252 in 2024 (nothing inferred).
- Snowmobiles in Québec: 173,159 in 2011, 237,201 in 2022 (published), 229,497 in 2024 (inferred).
- Mopeds in Québec: 34,041 in 2011, 37,511 in 2022 (published), 36,831 in 2024 (inferred).



# The 2011–2021 trend carried forward

**Montréal: the 2011–2021 trend carried forward, and what the files show**

Automobiles and light trucks, municipality of Montréal. The axis does not start at zero.

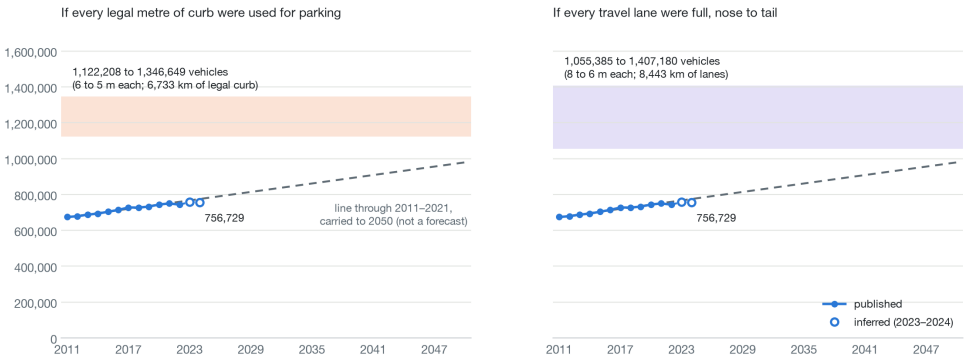



The line is not a forecast: it shows where the counts of 2011–2021 were heading. 2022 is the published count; 2023 and 2024 are inferred from make, model and model year (within 0.13 % on published years).
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# The same series against two hypothetical bounds

**Montréal's automobiles against two hypothetical bounds of its streets**

Automobiles and light trucks, municipality of Montréal; the bands are limits under stated assumptions, not measurements




Parking: City géobase (October 14, 2026), 8,257 km of curb without autoroutes, bridges, tunnels, ramps and back lanes; less 5 m at each end of each block face, and less the stretches where the City's sign inventory (October 23, 2026) forbids stopping or parking at all times (10.9 % in the boroughs where the inventory is complete, applied to the others). Driveways, hydrants, bus stops and hourly or permit restrictions are not deducted. Driving: OpenStreetMap (Geofabrik, November 11, 2026), autoroutes included, lane counts from the map for 86 % of lane-kilometres. Vehicles kept off the street are in neither bound; the registered vehicles are never all on the road at once; vehicles from elsewhere are not counted; traffic stops flowing well before every lane is full. © OpenStreetMap contributors.
Sources: SAAQ, open data « Véhicules en circulation », 2011–2022 (files of December 18, 2023); SAAQ, files released under the access Act, file 03.06.34729 (decision of September 17, 2025), 2023–2024.

# Vehicles and the space they occupy

| | Vehicles | Curb, if all on the street | |
|---|---|---|---|
| | | at 5.0 m each | at 6.0 m each |
| Montréal: published increase 2011 to 2021 | 75,574 | 377.9 km | 453.4 km |
| Montréal: 2024 (inferred) minus 2021 | 6,658 | 33.3 km | 39.9 km |
| Montréal: trend 2011-2021 carried from 2024 to 2034 (10 years) | 78,546 | 392.7 km | 471.3 km |
| H4A and H4B, if the City's 2011-2021 yearly rate applied for 10 years | 1,869 | 9.3 km | 11.2 km |
| Per 1,000 vehicles | 1,000 | 5.0 km | 6.0 km |

- The length per vehicle is an assumption given as a range, and the files do not say which vehicles are kept on the street. The table converts a number of vehicles into a length; it does not measure parking.
- Per 1,000 residents of the municipality of Montréal: 419 automobiles and light trucks at the 2016 Census, 426 at the 2021 Census.
- The line through 2011–2021 adds 7,855 a year. The 2024 count (inferred) is 18,213 below it.

# Limits, and what is not yet done

- Type and fuel for 2023–2024 are inferred; class cannot be. Electric and plug-in counts outside 2017–2022 are inferred and shown as such; forward they run low (electric by 4 to 19 % on published years).
- Mass and licences stop at 2022. By FSA there are two years, so no local trend.
- Registrations count vehicles owned, not trips made or distance driven. Vehicles owned by businesses are included everywhere.
- The reason for the lower published count of 2022 is not established here.
- Not yet done: electric vehicles for 2023–2024 with an outside reference for new models; yearly population estimates; the French versions of the note and deck.
- Code, tables and figures: `github.com/OpenMobilityData/Quebec-vehicle-registrations`